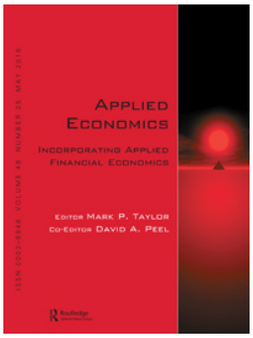

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A discrete time model of convergence for the term structure of interest rates in the case of entering a monetary union

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ABSTRACT
This paper presents a method for constructing the term structure of interest rate spreads for two currencies in the context of a country's entry into a monetary union. We propose a special type of process that ensures the convergence of the short-term interest rate spread to zero by a fixed moment in time, which we call the discrete-time Brownian bridge process. Using this process and the conventional pricing kernel framework, we derive double recursive formulas for computing the affine coefficients for the term structure of interest rate spread. The estimated model counterpart, which is based on the pre-EMU interest rate spread data for the interest rates of the German mark and Italian lira, fits the data reasonably well and captures the stylized empirical facts. Namely, spreads for all maturities have downward trends, and the longer the maturity is, the less spread there is.

KEYWORDS
Pricing kernel; monetary union; Brownian bridge; yield curve

JEL CLASSIFICATION
E43; G15; G12; F33

I. Introduction

For international term structure models, the yield spread between different countries is determined by exchange rate risk. After joining a currency area, this yield spread should disappear.[1] Such future events certainly affect the current prices of financial assets. In this paper, we develop a term structure model of interest rates for a country that will join a currency area in the future. For this purpose, we first choose the discrete-time pricing kernel modelling framework (Backus, Foresi, and Telmer 2000) for the term structure of interest rate spread. Second, to specify short-term interest rate spread dynamics, we introduce a special process, which we refer to as the discrete-time Brownian bridge process. This stochastic process converges to zero by a specific moment in time. Therefore, this specification ensures the convergence of the short-term spread to zero by the country's entrance into the currency area, thereby preventing the emergence of arbitrage opportunities at that time.

We compute the term structure of interest rate spreads using a recurrent procedure. Unlike Backus, Foresi, and Telmer (1998, 2000), we employ two inductions because the affine coefficients now depend on time. To fit the model to the data, we use money market and swap rate spreads between Italian and German interest rates for the time before the introduction of the euro. As an econometric approach, we choose maximum-likelihood estimation, assuming that the one-month spread is fully observable. The estimation outcome shows a reasonably good fit of the model to the data. The presented model may be of interest to groups of countries planning to establish a monetary union, such as the East African Community, the Association of Southeast Asian Nations (ASEAN) and the Cooperation Council for the Arab States of the Gulf.

Most papers in the field are devoted to developing methods to estimate the probability of EMU entry for EU member states from market data before the euro project began, such as De Grauwe (1996), Favero et al. (2000) and Bates (1999). Lund (1999) analyses the effect of planned EMU membership on domestic yield curves in potential member states. To describe the dynamics of the short-term spread (the spread between domestic and euro interest rates), Lund applies the Vasicek process (or, in other words, the Ornstein–Uhlenbeck process). It is not difficult to

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[1]We assume that differences in factors such as credit risk, liquidity, and taxes do not exist. The differences in sovereign spreads within a currency area has of course been a topic of great interest in recent years; however, it is not our purpose to study this phenomenon here.

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prove that, to prevent arbitrage opportunities, the short-term spread should converge to zero by the time the currency area comes into effect. Lund's specification does not ensure the fulfilment of this condition. Santamaria and Schwartz (2000) consider a model in which the domestic interest rate converges asymptotically to the 'European rate'; however, in their specification, the interest rate spread does not disappear after the beginning of the euro project.

The rest of the paper is organized as follows. Section II presents the model. Section III presents the econometric approach and the empirical results. Section IV concludes the paper.

II. The model

Without loss of generality, the Eurozone (EMU) can be used as an example of a monetary union, with the euro being a common currency. The currency of a country planning to enter the Eurozone is called the domestic currency. If this country joins the EMU at time T, future nominal claims are redenominated to euros at the exchange rate at time T. Furthermore, assume that the spread between the euro and domestic short-term nominal interest rates is zero at time T. The reason for this assumption is that it guarantees equal domestic and euro interest rates on the date of entrance into the Eurozone; otherwise, arbitrage would be possible. Indeed, let us borrow for an infinitesimal term before the entrance in currency with a lower short-term interest rate, convert it into another currency and lend it for the same term. At the same time, both currencies merge. However, the difference between deposit and loan rates remains, which gives us an arbitrage profit.

To provide the convergence of the short-term interest rate spread, z_t, to zero by the date of entrance into a monetary union, we define its dynamics by the following stochastic difference equation:

$$z_{t+1} = \left(1 - \frac{1}{T-t}\right) z_t + \sigma \varepsilon_{t+1}, \tag{1}$$

where T is the time of entrance into the monetary union, ε_t are normally distributed random disturbances with zero mean and unit variance, and σ is the conditional volatility of the process. We refer to the process (1) as a discrete-time Brownian motion process. This specification is a discrete-time version of the conventional continuous-time Brownian bridge process,

$$dx_t = -\frac{x_t}{T-t} dt + \sigma dw_t,$$

where w_t is the standard Wiener process. It is known that the process x_t converges to zero almost surely (a.s.) as $t \to T$ (see e.g. Karatzas and Shreve 1991; Protter 2005). The same feature is also valid for process (1), i.e. it is equal to zero at time T if $\varepsilon_T = 0$ a.s. Indeed, substituting $T-1$ for t in (1) gives

$$\begin{aligned} z_T &= \left(1 - \frac{1}{T-T+1}\right) z_{T-1} + \sigma \varepsilon_T \\ &= (1-1) z_{T-1} + 0 = 0. \end{aligned}$$

We also assume that $z_t = 0$ and $\varepsilon_t = 0$ if $t > T$.

The short-term interest rate in the domestic country is $r_t = R_t + z_t$, where R_t is the risk-free short-term interest rate in the EMU. We next assume that R_t is a function of a vector of state variables Z_t and that the state variables are governed by the following stochastic difference equation:

$$Z_{t+1} = \mu(Z_t) + G(Z_t)\Sigma_{t+1},$$

where $\mu(Z_t)$ and $G(Z_t)$ are mappings, which can be linear, and Σ_{t+1} is a vector of independent disturbances that is also independent of ε_t for all t. Let $m_{e,\ t} = m_e(Z_{t-1}, \Sigma_t)$ be the pricing kernel for the euro bond pricing (Backus, Foresi, and Telmer 1998, 2000); assume that it does not depend on the domestic factor and shock z_t and ε_t. Suppose that the log of the pricing kernel for the domestic bond pricing has the following form:

$$\log(m_{d,t}) = \log(m_{e,t}) + \log(m_t), \tag{2}$$

where $m_t = m(z_{t-1}, \varepsilon_t)$ is a function of z_{t-1} and ε_t only. The random variables $m_{e,\ t}$ and m_t are obviously independent.

The price of the $(n+1)$-period zero-coupon domestic bond, then, is

$$b_t^{n+1} = E_t(m_{d,t+1} b_{t+1}^n), \tag{3}$$

where E_t is the conditional expectations at time t (Backus, Foresi, and Telmer 1998, 2000). Using (2) and (3) gives us

$$\begin{aligned} b_t^{n+1} &= E_t\left[(m_{e,t+1} m_t) b_{t+1}^n\right] \\ &= E_t\left[(m_{e,t+1} m_t) \exp[-(\delta_{t+1}^n n + y_{t+1}^n n)]\right] \\ &= E_t\left[(m_{e,t+1} m_t) \exp(-\delta_{t+1}^n n) \exp(-y_{t+1}^n n)\right], \end{aligned} \tag{4}$$

where y_{t+1}^n and δ_{t+1}^n are the euro interest rate and spread of a maturity n, respectively; therefore,

$\delta^n_{t+1}+y^n_{t+1}$ is the domestic interest rate of maturity n. If ε_t and Σ_t are independent, then it easily follows that δ^n_{t+1} and y^n_{t+1} are also independent for all t. From this and (4), it follows that

$$b^{n+1}_t = E_t\left[m_{e,t+1}\exp(-y^n_{t+1}n)\right]\cdot E_t\left[m_{t+1}\exp(-\delta^n_{t+1}n)\right] = b^{n+1}_{e,t}d^{n+1}_t,$$

where $b^{n+1}_{e,t} = E_t\left[m_{e,t+1}\exp(-y^n_{t+1}n)\right]$ is the euro bond price and $d^{n+1}_t = E_t\left[m_{t+1}\exp(-\delta^n_{t+1}n)\right]$ is a domestic discount factor. Because our focus is primarily on the term structure of spread, in what follows, we shall ignore the issues related to the euro interest rates. Taking into account that $d^n_t = \exp(-\delta^n_t n)$ and $d^0_t = 1$, we can treat the domestic discount factor as a 'bond', which, however, can take values greater than one in the case in which spreads are negative. At the same time, the function $m_t = m(z_{t-1}, \varepsilon_t)$ is a pricing kernel related to this type of 'bond'.

We now derive the domestic discount factor for various maturities. Taking into account an analogy between the domestic discount factors and price of bonds, we follow Backus, Foresi, and Telmer (2000). We define the pricing kernel of the domestic discount factor as

$$-\log m_{t+1} = \begin{cases} \lambda^2/2 + z_t + \lambda\varepsilon_{t+1}; t = 0, 1, 2, \ldots, T-1, \\ z_t; t = T-1, \\ 0; t \geq T \end{cases} \tag{5}$$

where $\lambda$is the price of risk for the spread. This specification guarantees that the spread risk disappears for the times $t \geq T-1$. The $(n+1)$-period domestic discount factor is

$$d^{n+1}_t = E_t(m_{t+1} \times d^n_{t+1}). \tag{6}$$

Assume that the n-period domestic discount factor can be expressed as

$$-\log d^n_t = A_{n,t} + B_{n,t}z_t, \tag{7}$$

where the parameters $A_{n,t}$, $B_{n,t}$; $n = 0, 1, 2, \ldots, T-t$; $t = 0, 1, 2, \ldots, T-1$ ought to be calculated.

We note that, as opposed to the model of Backus, Foresi, and Telmer (2000), the coefficients $A_{n,t}$ and $B_{n,t}$ are not time invariant. The domestic discount factors can be derived by double induction. Indeed, from the initial condition $d^0_t = 1$, it follows that $A_{0,t} = B_{0,t} = 0$, $(t = 0, 1, 2, \ldots, T-1)$. From the log-normal property of m_{t+1}, the condition $d^0_t = 1$ and (6), it easily follows that $-\log d^1_t = z_t$. This means that for the one-period discount factor, we have $A_{1,t} = 0$ and $B_{1,t} = 1$, $(t = 0, 1, 2, \ldots, T-1)$. We can now begin the induction process. Given the coefficients $\{A_{n,t}, B_{n,t}, t = 0, 1, 2, \ldots, T-1, n \leq T-t\}$ and using (5) and (7), we may derive formulae for the coefficients $\{A_{n+1,t}, B_{n+1,t}\ (t = 0, 1, 2, \ldots, T-1)\}$. The log of the term under expectations in (6) is

$$\begin{aligned}
\log m_{t+1}d^n_{t+1} &= \log m_{t+1} + \log d^n_{t+1} \\
&= -\lambda^2/2 - z_t - \lambda\varepsilon_{t+1} \\
&\quad - A_{n,t+1} - B_{n,t+1}z_{t+1} \\
&= -\lambda^2/2 - z_t - \lambda\varepsilon_{t+1} - A_{n,t+1} \\
&\quad - B_{n,t+1}\left[\left(1 - \frac{1}{T-t}\right)z_t + \sigma\varepsilon_{t+1}\right] \\
&= -\left[\lambda_1^2/2 + A_{n,t+1}\right] \\
&\quad - \left[1 + B_{n,t+1}\left(1 - \frac{1}{T-t}\right)\right]z_t \\
&\quad - \left(\lambda_1 + B_{n,t+1}\sigma\right)\varepsilon_{t+1},
\end{aligned}$$

with mean

$$\begin{aligned}
E_t\left(\log m_{t+1} + \log d^n_{t+1}\right) = &-\left[\lambda_1^2/2 + A_{n,t+1}\right] \\
&- \left[1 + B_{n,1,1}\left(1 - \frac{1}{T-t}\right)\right]z_t
\end{aligned}$$

and variance

$$V_t\left(\log m_{t+1} + \log d^n_{t+1}\right) = \left(\lambda_1 + B_{n,t+1}\sigma_1\right)^2.$$

Using (1), the property of the log-normal distribution and the mean and variance of $\log(m_{t+1}d^n_{t+1})$, as computed above, yields

$$\begin{aligned}
-\log d^{n+1}_t = &-\log E(m_{t+1}d^n_{t+1}) = \left[\lambda^2/2 + A_{n,t+1}\right] \\
&+ \left[1 + B_{n,t+1}k\left(1 - \frac{1}{T-t}\right)\right]z_t \\
&- \frac{\left(\lambda_1 + B_{n,t+1}\sigma\right)^2}{2} \\
= &\underbrace{\left[\lambda_1^2/2 + A_{n,t+1}\right] - \frac{\left(\lambda + B_{n,t+1}\sigma\right)^2}{2}}_{A_{n+1,t}} \\
&+ \underbrace{\left[1 + B_{n,t+1}\left(1 - \frac{1}{T-t}\right)\right]}_{B_{n+1,t}} z_t \\
= &A_{n+1,t} + B_{n+1,t}z_t.
\end{aligned}$$

Finally, we obtain the following recursive relationships:

$$A_{n+1,t} = A_{n,t+1} + \lambda^2/2 - \frac{\left(\lambda + B_{n,t+1}\sigma\right)^2}{2}, \tag{8}$$

$$B_{n+1,t} = 1 + B_{n,t+1}\left(1 - \frac{1}{T-t}\right), \tag{9}$$

if $n + 1 \le T\text{–}t$ and $t \le T\text{–}1$.

Equation 8 implies that the value of $A_{n,\ t+1}$ is necessary to compute $A_{n+1,t}$. As a result, we may also compute the parameters $A_{n,\ t+1}$ recursively by t, taking into account that at a fixed t, the index n takes values $1, 2, \ldots, T\text{–}t$. In other words, we start with $A_{1,T-1} = 0$, then by using (8) and $B_{1,T-1}$, which is known from (9), we compute $A_{2,T-2}$ and, in the same way, $A_{2,T-3}$, $A_{3,T-3}$, and so on. Because the spread disappears for $t \ge T$, it is easily seen that

$$A_{n,t} = 0 \text{ and } B_{n,t} = 0, \text{for all } n \ge 0 \text{ and } t \ge T. \tag{10}$$

For $t = T\text{–}1$ from (6), it follows that $d_{T-1}^{n+1} = E_{T-1}(m_T d_T^n)$. Then, from (5) and the condition $\varepsilon_T = 0$, we have

$$\log m_T d_T^n = \log m_T + \log d_T^n = -z_{T-1}. \tag{11}$$

From (8) to (10), we obtain

$$B_{n+1,T-1} = 1 \textit{ and } A_{n+1,T-1} = A_{n,T} = 0, n = 0, 1. \tag{12}$$

For the time $t = T\text{–}2$ using (5)–(7) and (8)–(9), we obtain

$$A_{n+1,T-2} = \begin{cases} \frac{\lambda^2}{2} + A_{n,T-1} - \frac{\left(\lambda + B_{n,T-1}\sigma\right)^2}{2} = \frac{\lambda^2}{2} - \frac{(\lambda+\sigma)^2}{2} = -\frac{(2\lambda+\sigma)\sigma}{2}; n > 0 \\ 0; n = 0 \end{cases}, \tag{13}$$

$$B_{n+1,T-2} = \begin{cases} 1 + B_{n,T-1}\left(1 - \frac{1}{T-T+2}\right) = \frac{3}{2}; n > 0 \\ 1; n = 0 \end{cases}. \tag{14}$$

Note that for fixed t, the coefficients $A_{n,\ t}$ are equal for all maturities n longer than $T\text{–}t$. The same is valid for $B_{n,\ t}$. Proceeding with (8) and (9) backward in t, we see that this is true for all times $0 < t \le T$.

Let us rewrite (9) in the form

$$B_{n,t} = 1 + B_{n-1,t+1}\frac{T-t-1}{T-t}. \tag{15}$$

For $n \le T\text{–}t$, iterating (15) forward with respect to the first subscript of B, we have

$$\begin{aligned} B_{n,t} &= 1 + \left(1 + B_{n-2,t+2}\left(\frac{T-t-2}{T-t-1}\right)\right)\frac{T-t-1}{T-t} \\ &= 1 + \left(\frac{T-t-1}{T-t}\right) + \left(\frac{T-t-1}{T-t}\right)\left(\frac{T-t-2}{T-t-1}\right) \\ &\quad + \cdots + \left(\frac{T-t-1}{T-t}\right)\left(\frac{T-t-2}{T-t-1}\right)\left(\frac{T-t-3}{T-t-2}\right) \\ &\quad \cdots \left(\frac{T-t-n+1}{T-t-n+2}\right). \end{aligned}$$

This can be simplified as follows:

$$\begin{aligned} B_{n,t} &= 1 + \left(\frac{T-t-1}{T-t}\right) + \left(\frac{T-t-2}{T-t}\right) \\ &\quad + \left(\frac{T-t-3}{T-t}\right) + \ldots + \left(\frac{T-t-n+1}{T-t}\right) \\ &= 1 + \left(1 - \frac{1}{T-t}\right) + \left(1 - \frac{2}{T-t}\right) \\ &\quad + \left(1 - \frac{3}{T-t}\right) + \ldots + \left(1 - \frac{n-1}{T-t}\right) \\ &= n - \frac{1}{T-t}(1 + 2 + 3 + \ldots + (n-1)) \\ &= n - \frac{(n-1)n}{2(T-t)} = n\left(1 - \frac{n-1}{2(T-t)}\right). \end{aligned} \tag{16}$$

For $n > T\text{–}t$, it follows that

$$\begin{aligned} B_{n,t} &= T - t - \frac{(T-t-1)(T-t)}{2(T-t)} \\ &= T - t - \frac{(T-t-1)}{2} = \frac{T-t+1}{2}. \end{aligned} \tag{17}$$

Equations (16) and (17) can be rewritten as

$$B_{n,t} = \begin{cases} n\left(1 - \frac{(n-1)}{2(T-t)}\right), n \le T - t; t = 0, 1, \ldots T-1, \\ \frac{1}{2(T-t+1)}, n > T - t; t = 0, 1, \ldots T-1 \end{cases}. \tag{18}$$

Taking into account (8) and (12), we can compute the coefficients $A_{n,\ t+1}$ for $n > T\text{–}t$ in the same way, i.e.

$$
\begin{aligned}
A_{n+1,t} &= A_{n,t+1} + \frac{\lambda^2}{2} - \frac{1}{2}\left(\lambda + \frac{T-t}{2}\sigma\right)^2 \\
&= A_{n-1,t+2} + \frac{\lambda^2}{2} - \frac{1}{2}\left(\lambda + \frac{T-t-1}{2}\sigma\right)^2 \\
&\quad + \frac{\lambda^2}{2} - \frac{1}{2}\left(\lambda + \frac{T-t}{2}\sigma\right)^2 \\
&= (T-t)\frac{\lambda^2}{2} - \frac{1}{2}\sum_{\tau=0}^{T-1}\left(\lambda + \frac{T-t-\tau}{2}\sigma\right)^2.
\end{aligned}
$$

Finally, spreads are expressed as a linear function of the state variable z:

$$\delta_t^n = -\frac{1}{n}\log d_t^n = \frac{1}{n}\left(A_{n,t} + B_{n,t}z_t\right). \tag{19}$$

III. Econometric approach and empirical results

Data

We now compare the outcome of the proposed model with the data. As the case study, we choose the situation preceding the introduction of the euro as a single European currency. On 1 January 1999, 11 European countries abandoned their national currencies and began using the single currency in their territories. After joining the EMU, the yield spread between the euro area government bonds should disappear.

To estimate the model parameters (λ, σ), we use money market and swap rate spreads for the German mark and the Italian lira. The money market data include rates with maturities of *1, 3, 6* and *12* months, whereas the swap market data include rates with maturities of *2, 3, 4* and *5* years. The yields are sampled weekly (Wednesdays) from 20 August 1997 to 12 August 1998, i.e. overall $N = 52$.

Figure 1 illustrates the behaviour of the historical spreads of various maturities for the period under consideration. The noticeable features of the data are as follows: (i) spreads for all maturities have downward trends and (ii) the longer maturity is, the less spread there is. We will show that the proposed model is able to capture both of these features.

Estimation strategy and discussion

We treat the weekly rate as an unobservable factor z_t and assume that the one-month rate is observed without errors, whereas all the other yields include

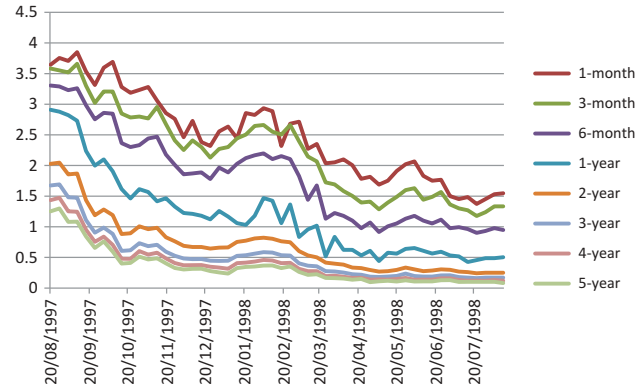

Figure 1. Interest rate spreads of the Italian lira and German mark for various maturities in percentage points.

measurement errors. Taking this into account, using (19), we can extract the unobservable factor z_t from the one-month rate

$$z_t = \left(B_{4,t}\right)^{-1}\left(4\delta_t^4 - A_{4,t}\right).$$

Putting this into (19), for other yields, we have

$$\delta_t^n = \frac{1}{n}\left(A_{n,t} + B_{n,t}\left(B_{4,t}\right)^{-1}\left(4\delta_t^4 - A_{4,t}\right)\right) + \eta_t^n,$$

where η_t^n are measurement errors for the spread of maturity n. We choose a cross-sectional econometric approach (see e.g. Chen and Scott 1993; Brown and Schaefer 1994; Duffie and Singleton 1997), i.e. we estimate the parameters by using cross-sectional information (yields for different maturities). We assume that the standard deviations of the measurement errors for all maturities are equal. Then, maximizing the likelihood function amounts to minimize the sum of the squared errors of all 'unobserved' yields,

$$\arg\ \max(\ln L) = \arg\ \min \sum_{t=1}^{N}\sum_{n\in u}\left(\eta_t^n\right)^2, \tag{20}$$

where u is a set of the 'unobserved' maturities. Using the Kalman filter approach as in Duan and Simonato (1999) and Babbs and Nowman (1999), we encounter the problem of parameter identification, whereas our proposed econometric approach guarantees a unique solution to (20). This is another advantage of our approach over the Kalman filter approach.

We only need to estimate two parameters: the price of risk λ and the volatility σ of the process z_t. The estimated value of λ is 23.9 with a standard error of 1.8 annual percentage points, and the

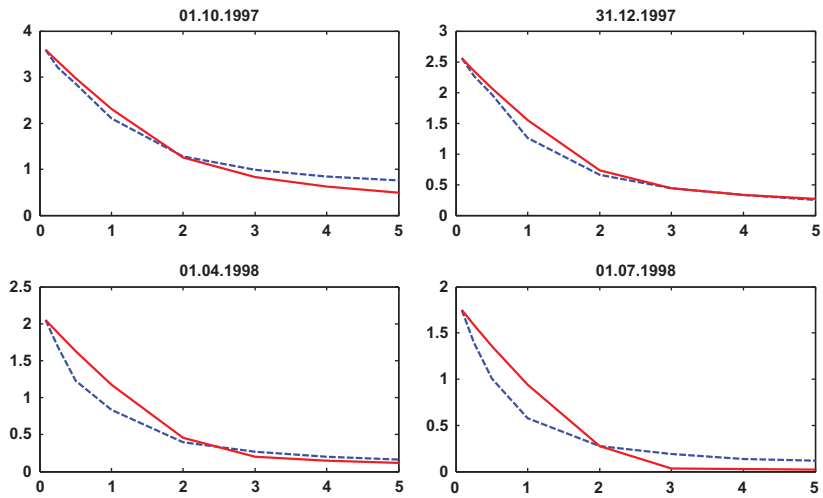

Figure 2. Model and data term structures of spreads between Italian lira and German mark interest rates for different dates. Notes: The red line is the model term structure of spread; the blue line is the actual data of the term structure of spreads. The horizontal axis is maturity in years.

estimate of σ is 2.0×10^{-7} with a standard error of 1.5×10^{-8}. These estimation results seem reasonable.

Figure 2 displays the theoretical and actual term structures of spreads for the dates of 01.10.1997, 31.12.1997, 01.04.1998 and 01.07.1998, which correspond to five-, four-, three- and two-quarter terms before the date of the euro's introduction, respectively. The figure shows that the model fits the data fairly well. The maximum differences in the term of spreads between the model and data are 0.26, 0.10, 0.40 and 0.36 percentage points for the dates 01.10.1997, 31.12.1997, 01.04.1998 and 01.07.1998, respectively.

Figure 3 illustrates the dynamics of the computed theoretical spreads under the estimated coefficients, i.e. one-step model forecasts and actual spreads. For the short maturities of up to one year, we have fairly good approximations for the data. For the long maturities, 2–5 years, the model also approximates the data well after approximately 10 weeks from the beginning of the sample. However, the deviation of the model time series from the data is significant at the beginning of the sample. This deviation can be explained by the fact that at the beginning of the considered period, August 1997, market participants might not have been sure that Italy would enter the Eurozone, and thus, the short-term spread might not have followed the Brownian bridge process at that time. Another explanation may be that the spread term structure is driven by more than one factor (especially at the beginning of the considered period), as is usual in term structure interest rate models, where additional factors significantly improve the fit.[2]

IV. Conclusions

We propose a model for deriving the term structure of interest rate spreads in the case of a country's entrance into a monetary union. For this purpose, we introduce a new type of stochastic process called the discrete-time Brownian bridge process. This process allows us to construct a consistent model of the term structure of interest rate spreads by preventing the emergence of arbitrage opportunities at the time when a country enters a monetary union. To check the empirical quality of the model, we apply pre-EMU data for the interest rate spreads between the Italian lira and the German mark. To estimate the model, we use the maximum-likelihood approach, and estimation results show a fairly good model fit to the data, except for the very beginning of the period under consideration. Additionally, the advantage of the model is its ability to reproduce stylized features of the data, namely: (i) spreads for all maturities have downward trends and (ii) the longer the maturity is, the less spread there is.

[2]Because the sample under consideration is already somewhat small, reducing it further would be undesirable for the quality of the estimation.

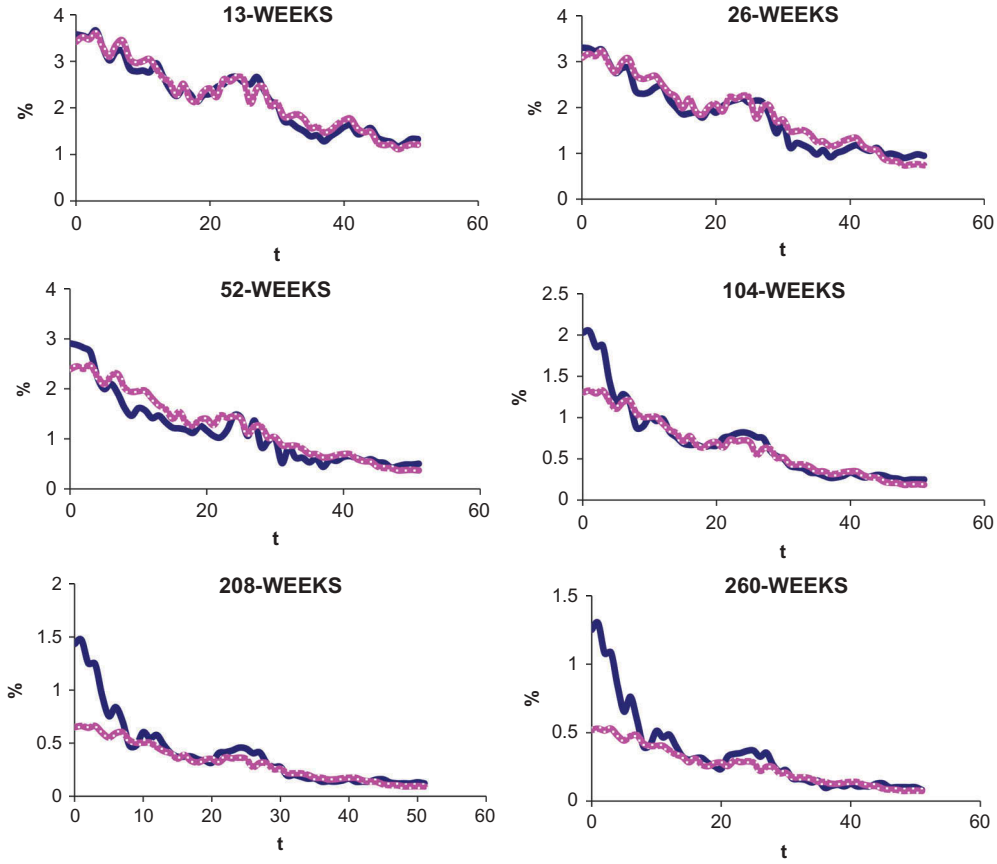

Figure 3. Observable and calculated spreads between Italian lira and German mark interest rates for maturities of 13, 26, 52, 104, 208, and 260 weeks. Notes: The purple line is the model spreads; the blue line is actual data over the data sample. The horizontal axis is time in weeks from 20 August 1997.

The application of the model to the evaluation of fixed-income instruments may give more correct prices for the instruments in the case of a country's entrance into a monetary union and may help create hedging strategies and fixed-income portfolios. Generally, the proposed method can be used to extract the part of the yield spread relating to the exchange rate risk. The rest of the spread may be attributed to credit risk, liquidity risk, or other factors. Using a multi-dimensional version of the discrete-time Brownian bridge process might improve the fit of models to the data; however, the authors leave this to future research.

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